

BMATS101 – MATHEMATICS-I (CSE)

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MODULE 4

MODULAR ARITHMETIC

CONGRUENCES & BASIC NUMBER THEORY

1. Introduction to Modular Arithmetic

Modular arithmetic is a system of arithmetic for integers where numbers **wrap around after reaching a certain value**, called the modulus.

It forms the mathematical foundation for:

- Cryptography
- Computer algorithms
- Coding and decoding
- Data security

Modular arithmetic is widely used in **RSA encryption** and other public key systems.

2. Congruences

2.1 Definition

Let a , b , and m be integers with $m > 0$.

We say that a is **congruent to b modulo m** if their difference is divisible by m .

$$a \equiv b \pmod{m}$$

This means:

$$m \mid (a - b)$$

2.2 Examples

$$17 \equiv 5 \pmod{12}$$

because $17 - 5 = 12$

$$-3 \equiv 2(\text{mod}5)$$

because $-3 - 2 = -5$

3. Properties of Congruences

Let:

$$a \equiv b(\text{mod}m), c \equiv d(\text{mod}m)$$

Then,

Addition:

$$a + c \equiv b + d(\text{mod}m)$$

Subtraction:

$$a - c \equiv b - d(\text{mod}m)$$

Multiplication:

$$ac \equiv bd(\text{mod}m)$$

Exponentiation:

$$a^n \equiv b^n(\text{mod}m)$$

4. Linear Congruences

4.1 Definition

A congruence of the form:

$$ax \equiv b(\text{mod}m)$$

is called a **linear congruence**.

4.2 Condition for Existence of Solution

The congruence $ax \equiv b(\text{mod}m)$ has a solution **if and only if**:

$$\text{gcd}(a, m) \mid b$$

5. Solving Linear Congruences

Example 1

Solve:

$$3x \equiv 1(\text{mod}7)$$

Solution:

We look for a number x such that:

$$3x = 1 + 7k$$

Try values:

$$3 \cdot 5 = 15 \equiv 1(\text{mod}7)$$

Hence,

$$\boxed{x \equiv 5(\text{mod}7)}$$

Example 2

Solve:

$$\begin{aligned} 4x &\equiv 8(\text{mod}12) \\ \gcd(4,12) &= 4, 4 \mid 8 \end{aligned}$$

Divide by 4:

$$x \equiv 2(\text{mod}3)$$

6. System of Linear Congruences

A set of congruences such as:

$$\begin{aligned} x &\equiv a_1(\text{mod}m_1) \\ x &\equiv a_2(\text{mod}m_2) \end{aligned}$$

is called a **system of linear congruences**.

7. Chinese Remainder Theorem (CRT)

7.1 Statement

If $m_1, m_2, \dots, m_n$ are pairwise coprime integers, then the system:

$$\begin{aligned} x &\equiv a_1(\text{mod}m_1) \\ x &\equiv a_2(\text{mod}m_2) \\ &\vdots \\ x &\equiv a_n(\text{mod}m_n) \end{aligned}$$

has a **unique solution modulo** $m_1 m_2 \cdots m_n$.

Example 3

Solve:

$$x \equiv 1 \pmod{3}$$

$$x \equiv 2 \pmod{5}$$

Try values satisfying first congruence:

$$x = 1, 4, 7, 10, \dots$$

Check modulo 5:

$$7 \equiv 2 \pmod{5}$$

Hence:

$$x \equiv 7 \pmod{15}$$

8. Euler's Totient Function

8.1 Definition

Euler's totient function $\phi(n)$ is defined as the number of positive integers **less than or equal to n** that are **coprime to n** .

8.2 Values

If p is prime:

$$\phi(p) = p - 1$$

If $n = p^k$:

$$\phi(n) = p^k - p^{k-1}$$

If $n = p_1^{a_1} p_2^{a_2} \dots p_r^{a_r}$:

$$\phi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_r}\right)$$

Example 4

Find $\phi(20)$

$$20 = 2^2 \cdot 5$$

$$\phi(20) = 20 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right) = 8$$

EULER'S THEOREM, FERMAT'S LITTLE THEOREM, WILSON'S THEOREM & RSA ALGORITHM

1. Euler's Theorem

1.1 Statement

If a and n are **coprime positive integers**, then:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

where $\phi(n)$ is Euler's totient function.

1.2 Meaning

Euler's theorem generalizes Fermat's little theorem and is used to:

- Simplify large powers modulo n
- Form the mathematical basis of encryption algorithms

1.3 Example 1

Find:

$$3^{40} \pmod{20}$$

Solution:

$$\phi(20) = 8$$

Since $\gcd(3, 20) = 1$,

$$3^8 \equiv 1 \pmod{20}$$

$$3^{40} = (3^8)^5 \equiv 1^5 \equiv 1 \pmod{20}$$

2. Fermat's Little Theorem

2.1 Statement

If p is a **prime number** and a is an integer such that $p \nmid a$, then:

$$a^{p-1} \equiv 1 \pmod{p}$$

2.2 Alternative Form

$$a^p \equiv a \pmod{p}$$

2.3 Example 2

Find:

$$7^{100} \pmod{13}$$

Solution:

Since 13 is prime,

$$\begin{aligned}\phi(13) &= 12 \\ 7^{12} &\equiv 1(\text{mod}13) \\ 7^{100} &= 7^{96} \cdot 7^4 \equiv 1^8 \cdot 7^4 \\ 7^2 &= 49 \equiv 10, 7^4 \equiv 10^2 \equiv 9(\text{mod}13)\end{aligned}$$

3. Wilson's Theorem

3.1 Statement

A positive integer $p > 1$ is **prime if and only if**:

$$(p - 1)! \equiv -1(\text{mod}p)$$

3.2 Interpretation

Wilson's theorem provides a **test for primality**, though it is computationally expensive for large numbers.

3.3 Example 3

Verify Wilson's theorem for $p = 5$.

$$\begin{aligned}(5 - 1)! &= 24 \\ 24 &\equiv -1(\text{mod}5)\end{aligned}$$

Thus, Wilson's theorem is verified.

4. Comparison of Theorems

Euler's Theorem:

- Applies when $\text{gcd}(a, n) = 1$
- Works for any positive integer n

Fermat's Little Theorem:

- Special case of Euler's theorem
- Applies only when modulus is prime

Wilson's Theorem:

- Used as a primality test
- Uses factorial expression

5. Introduction to Cryptography

Cryptography is the science of **secure communication** using mathematical techniques.

Core goals:

- Confidentiality

- Authentication
- Integrity
- Non-repudiation

Modular arithmetic is the backbone of **public key cryptography**.

6. RSA Algorithm

6.1 Principle

RSA is a **public key encryption algorithm** based on:

- Large prime numbers
- Euler's theorem
- Modular exponentiation

6.2 Key Generation Steps

1. Choose two large primes p and q .
2. Compute:

$$n = pq$$

3. Compute:

$$\phi(n) = (p - 1)(q - 1)$$

4. Choose e such that:

$$\gcd(e, \phi(n)) = 1$$

5. Find d such that:

$$ed \equiv 1 \pmod{\phi(n)}$$

Public key: (e, n)

Private key: (d, n)

7. Encryption Process

Given message M :

$$C \equiv M^e \pmod{n}$$

where C is the ciphertext.

8. Decryption Process

$$M \equiv C^d \pmod{n}$$

This works due to Euler's theorem.

9. RSA Worked Example (Small Values)

Let:

$$p = 3, q = 11 \\ n = 33, \phi(n) = 20$$

Choose:

$$e = 3$$

Find d :

$$3d \equiv 1 \pmod{20} \Rightarrow d = 7$$

Public key: $(3, 33)$

Private key: $(7, 33)$

Encrypt $M = 4$:

$$C \equiv 4^3 = 64 \equiv 31 \pmod{33}$$

Decrypt:

$$M \equiv 31^7 \pmod{33} \Rightarrow 4$$

10. Applications of Modular Arithmetic

- Data encryption and decryption
- Secure internet communication
- Digital signatures
- Password protection
- Blockchain technology